

INCIDENT RESPONSE REPORT — Reconnaissance Phase (Windows 11 Target)

Report ID: IR-2026-002

Date of Incident: 2026-06-12

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Classification: Internal / Homelab Exercise

1. EXECUTIVE SUMMARY

On 2026-06-12, four network reconnaissance scans were conducted targeting Windows 11 host (192.168.80.131) from Kali Linux host (192.168.80.129) within an isolated lab environment. The objective of this test is to test Windows' endpoint detection against reconnaissance attempt. Despite having Sysmon and Windows security auditing deployed, Splunk SIEM detected 0 out of 4 reconnaissance activity. This identified a full blind spot for network layer reconnaissance on the Windows 11 host. The vuln script scan additionally confirmed that the host is hardened against SMB vulnerabilities.

2. INCIDENT DETAILS

Attack Phase: Reconnaissance (MITRE ATT&CK TA0043)

Technique: T1595 - Active Scanning

Source IP: 192.168.80.129 (Kali Linux)

Target IP: 192.168.80.131 (Windows 11)

Detection Source: Splunk Enterprise 10.4

LogSources: WinEventLog:Security, WinEventLog:System,

XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

Time Window: ~16:38 – 17:00 (Kali, UTC+8)

3. TIMELINE OF EVENTS

Scan Type: SYN Stealth | Command: nmap -sS | Detected: No | Notes: Sysmon network logging (EventID 3) disabled; half-open scan generates no completed-connection event

Scan Type: Version Detection | Command: nmap -sV | Detected: No | Notes: No reliable signature; EventID 4798/4799 events initially appeared correlated but were ruled out as background noise via controlled re-test

Scan Type: OS Detection | Command: nmap -O | Detected: No | Notes: No exact OS match returned; no log signature generated

Scan Type: Vuln Script | Command: nmap --script vuln | Detected: No | Notes: SMB negotiation rejected by host; no scan-related logs generated

4. ACTIONS CONDUCTED & DETECTION RESULTS

Action: SYN Stealth Scan | Command: nmap -sS 192.168.80.131 | Layer: TCP | Detected: No | Evidence: None — Sysmon network logging (EventID 3) disabled; half-open scan never completes a connection to log

Action: Version Detection | Command: nmap -sV 192.168.80.131 | Layer: Application | Detected: No | Evidence: None — suspected 4798/4799 correlation ruled out as background noise via controlled re-test

Action: OS Detection | Command: nmap -O 192.168.80.131 | Layer: Network/IP | Detected: No | Evidence: None — no exact OS fingerprint match, no log signature generated

Action: Vuln Script Scan | Command: nmap --script vuln 192.168.80.131 | Layer: Application |
Detected: No | Evidence: None — SMB negotiation rejected by host; no scan-related logs generated

5. FINDINGS & ANALYSIS

- Open Ports Discovered by Attacker:
 - 135/tcp — msrpc — Microsoft Windows RPC
 - 139/tcp — netbios-ssn — Microsoft Windows NetBIOS
 - 445/tcp — microsoft-ds — SMB
- Vulnerability Assessment Results:
 - smb-vuln-ms10-061: Connection could not be negotiated (not vulnerable)
 - samba-vuln-cve-2012-1182: Connection could not be negotiated (not vulnerable)
 - smb-vuln-ms10-054: false (not vulnerable)
 - The host rejected legacy SMB1 negotiation, indicating modern SMB hardening. The Windows 11 target is not susceptible to these classic SMB exploits.
- Key Observations:
 - All four scan types went completely undetected on the Windows host.
 - Sysmon EventID 3 (network connection logging) is disabled by the default olafhartong sysmon-modular configuration, removing visibility into network-layer scan activity.
 - A controlled re-test was performed to verify a suspected detection: EventID 4798/4799 (group membership enumeration) events initially appeared to correlate with scan times. However, a second scan produced no corresponding new events, confirming the original events were routine OS background activity and NOT scan-induced. This false positive was ruled out before being attributed as a detection.
 - The Windows host exhibits the same reconnaissance blind spot as the Ubuntu host (IR-2026-001), but for a different underlying reason: Ubuntu lacked network-layer logging entirely, while Windows has Sysmon deployed but with network logging disabled.

6. DETECTION GAP IDENTIFIED

The Windows endpoint has no reliable detection for network reconnaissance. Contributing factors:

- Sysmon EventID 3 (network connections) disabled by default config
- No network-based IDS deployed
- No Windows Firewall connection logging forwarded to Splunk

Reconnaissance scanning is currently undetectable across the entire lab environment (both hosts).

7. ANALYST NOTE

During analysis, EventID 4798/4799 enumeration events were observed near scan timestamps and initially appeared to be a detection. A controlled re-scan was performed to validate causation. The re-scan produced no new enumeration events, confirming the original events were coincidental OS background activity. This highlights the importance of correlation testing and baselining normal activity before attributing events as detections. Acting on the initial correlation alone would have produced a false-positive detection rule.

8. DETECTION QUERIES USED

Sysmon network connection check:

```
index=main host="DESKTOP*"
sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=3
"192.168.80.129"
```

Group enumeration check (false positive test):

```
index=main host="DESKTOP*" (EventCode=4798 OR EventCode=4799)
```

Combined attacker IP check:

```
index=main host="DESKTOP*" "192.168.80.129"
```

9. RECOMMENDATIONS

- Enable Sysmon EventID 3 (network connection logging), selectively scoped to listening ports (135/139/445) to balance visibility and noise.
- Deploy network-based IDS (Suricata or Zeek) to detect TCP/IP-layer scans that host logging cannot see.
- Enable Windows Firewall connection logging and forward to Splunk to capture connection attempts to closed ports.
- Build volume-based correlation rules (e.g., abnormal count of enumeration or connection events from a single source in a short window) rather than relying on individual event types.
- Baseline normal background activity before implementing alerts to avoid false positives such as the 4798/4799 events identified in this exercise.

10. MITRE ATT&CK MAPPING

Reconnaissance — Active Scanning — T1595

Reconnaissance — Gather Victim Host Information — T1592

11. CONCLUSION

The reconnaissance phase against the Windows target was entirely undetected by the Splunk SIEM, identifying a complete network-layer reconnaissance blind spot. While Sysmon and Security auditing are deployed, the default configuration does not provide network visibility. The exercise also validated, through controlled re-testing, that a suspected detection was in fact background noise — reinforcing the importance of analyst verification. The vuln scan confirmed

the host is hardened against legacy SMB exploits. Closing the reconnaissance detection gap will be addressed in a future iteration through network-layer monitoring and tuned correlation rules.

12. EVIDENCE / APPENDIX

END OF REPORT